

Review of MLib Validation

Market Risk & Methodology

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1 Validation Framework

This document is a review of the MHSC validation of MLib [MLibValMHSC17], with some supplementary analysis.

1.1 Scope

1.1.1 Products

Curves constructed from the following instruments: OIS, Central Bank Swap, Libor-OIS Basis Swap, Deposit Rates, FRA, Future, Swap, Tenor Basis Swap, Cross Currency Basis Swap

1.1.2 Model

MLib uses a bootstrapping approach by solving a sequence of Newton-Raphson optimizations for the discount factor at each quote maturity (pillar). For some calibration instruments (e.g. IRS), multi-variate Newton-Raphson is used.

Interpolation and state variable have default settings (same for all currencies). Interpolation method can be overridden by user, but the state variable is not exposed.

Collateralization is taken into account in the curve construction.

1.1.3 Implementation

MLib is written in C++ and is part of IR Server. It is accessible from Excel, Java, C#, R, and Python. We validated MLib via Excel Addin v20170915.2221, with the spreadsheet "20170523 CurveCalibrationtool-10.0.xlsb".

1.1.4 Use of the model

MLib is used in an unofficial capacity¹ in the MHI Front Office for curve construction and pricing. It is consumed by the systems Borat and Raptor. It is also used indirectly for the banking book.²

1.1.5 Validation scope

In this report we address the following two areas:

- Review of validation undertaken by MHSC
- Further analysis of zig-zag and curtain effects in OIS forward rates

1.2 Results and Conclusions

Model

MLib follows the industry standard approach to curve construction that accounts for collateralization, and allows flexibility in the choice of interpolation method. It is a conceptually sound approach.

Model documentation is lacking in details about the curve construction algorithm, but gives a good overview.

Implementation

The validation performed by MHSC demonstrates that MLib is a) internally consistent, and b) consistent with our benchmark library. For the former, calibration instruments are repriced in MLib with PV < 1bp for all but one case.³ For the latter, calibration instruments are repriced in a benchmark library (GMTLib) with PV < 1bp.

¹ Murex is currently used for EOD numbers.

² MHSC uses MLib for books & records pricing under the accrual method, and MHI adopts these prices when required.

³ Repricing the JPY 30Y Libor-OIS basis swap had an error of 1.3 bp. MHSC reported excellent agreement on par rates, with most diffs less than 1/1000th bp. For details, please consult [MLibValMHSC17].

Forward curves are generally smooth, except for OIS curves which can exhibit a zig-zag pattern on the short end, and a curtain effect in the medium / long end. After some analysis, we identified an incorrect curve setting and the zig-zag pattern has disappeared. The curtain effect is being addressed by the model owners.⁴

MLib relies on some legacy code (subject to on-going refactoring effort) that lacks transparency and provides maintenance challenges. This impacts the documentation and includes some hard coded features. The situation is improving as development continues. Use of MLib through Excel is convenient and intuitive, with good performance (speed).

Conclusion

We have no objection to the model (subject to the limitations below) except where mid / long end OIS forwards are calculated.

1.3 Main Limitations

1. **Materiality medium:** The OIS forwards can exhibit a curtain effect in the medium / long end of the curve
2. **Materiality low:** Curve setting *SmoothShortEnd = TRUE* introduces additional calibration points and is only intended with Central Bank Swaps in an OIS curve. This setting should be used with caution.
3. **Materiality low:** MLib documentation is lacking in some technical details (e.g. bootstrapping algorithm, Libor fixing).
4. **Materiality low:** Libor index fixing a) does not have a fixing lag, b) does not take a separate daycount (e.g. ACT/360). This is sufficient for the calibration instruments, but may not be for all trade pricing.
5. **Materiality low:** Interpolation state variables are hard-coded.
6. **Materiality low:** Libor-OIS bootstrapping strategy (back-to-back with Libor swaps) depends on the paired quotes having identical conventions on the float legs.
7. **Materiality low:** The par rate functions accuracy (~1/1000bp) is not consistent with the PV function accuracy (~1bp)

1.4 Performance Monitoring

If model use changes such that it is used for EOD numbers in the FO, we recommend the following check:

- Quarterly checking that quoted instruments are repriced to par (internal consistency)
- Quarterly monitoring of the smoothness of forward rates for traded products

1.5 Effective Challenge and Model Changes

While the model satisfied the validation tasks of demonstrating internal and external consistency, the shape of OIS forward rates prompted further investigation. As part of our review of the MHSC validation, we performed additional analysis on the forward rates and found that a) the effects can be isolated to the forward rate calculation, and b) the zig-zag issue on the short end is caused by a curve setting (*SmoothShortEnd = TRUE*). Model owners have recognized the curtain-effect, and the fix is to be validated at the next model review.

⁴ At the time of committee, a fix has been implemented that performs a global optimization on the OIS curve, and it will be validated at the next review

1.6 Mitigating actions

- The model should be reviewed again once the curtain effect fix is in place, and the anticipated dual-curve bootstrapping algorithm is implemented.
- The model should be reviewed if additional interpolation methods are introduced.

DRAFT

2 Model Review

Relying on the validation performed by MHSC in [MLibValMHSC17], we review the curve construction methodology and the numerical testing of the implementation. Then, we extend the original work with further analysis of OIS forwards.

2.1 Methodology

The bootstrapping approach is common in the industry. A sequence of Newton-Raphson optimizations is performed to bootstrap discount factors (one-to-one mapping with zero-rate) at each quote maturity. These represent the pillars. An interpolation method is assumed to obtain discount factors between pillars. Details for particular curves, and calibration instruments, are discussed in the MLib documentation [MLib17].

Some technical details of the bootstrap algorithm are not available. It is not clear how the Jacobian term is computed, or what is used for the initial guess. It is not clear how intermediate discount factors (between pillars) are calculated. In the case of a global interpolation scheme like cubic spline, each additional pillar will impact interpolation in the previous regions, so it may be difficult to solve for pillars consistently with the final curve.

2.2 Implementation

The basic requirement for curve validation is that it demonstrates internal consistency, by repricing quoted instruments to par. This is demonstrated in [MLibValMHSC17] by using curves bootstrapped in MLib to re-price the quoted instruments, giving a round trip error within tolerance. Of the instruments repriced, only the JPY Libor-OIS 30Y basis swap PV is greater than 1bp.⁵ Par rates are recalculated with excellent agreement, with most differences less than 1/1000 bp. The ON rate for the GBP OIS curve has the largest difference at 0.18bp. Please consult the MHSC validation for details.

The validation exercise goes beyond the basic requirement by using curves bootstrapped in MLib to reprice in an external library (GMTLib). All instruments re-price to par, within 1bp. A separate par rate calculation was not available in the benchmark library.

There is disparity between the MLib par rate ($\sim 1/1000$ bp diffs) and PV (~ 1 bp diffs) functions' accuracy.⁶ While it seems the par rate is the quantity of preference for traders, the user should be aware that the PV function is not on the same order of accuracy.

2.3 Notes on Design

MLib is emerging as a global curve construction solution. It is based on IR Server (Tokyo) code, which includes some legacy features. MLib is continuously improving through on-going development efforts. We note the following (whether by design, or legacy) that users should be aware of:

- Interpolation state variables are hard-coded for each curve type (i.e. OIS, Libor, Basis / Xccy curves)
- Libor Index fixing lag is not supported
- Libor fixing daycount (e.g. ACT/360) taken from float leg (not set independently)

In MLib, the state variable for a curve is a modelling choice. Other systems (e.g. Murex) make this configurable, as with the interpolation scheme, but it is fixed by curve type in MLib. It is not clear what methodology is behind the choice of state variable.

The latter two (Libor fixing) observations are not a problem for calibration since quoted instruments (currently) do not specify fixing lag, or a specific daycount. It may be a minor problem for pricing more

⁵ When reporting differences in bps, the absolute difference is normalized by notional so that the size of notional should have absolutely no impact.

⁶ Since par rates are annualized, to compare apples and apples the PV diffs should be annualized by dividing by the maturity. Even so, this is at best $1/30^{\text{th}}$ bp compared to $1/1000^{\text{th}}$ bp.

complex instruments. The Libor fixing daycount is absent from the swap pricing template (see appendix A). These issues may be related to dependency on legacy code, and can be understandably complicated to resolve. The Libor fixing procedure is outlined in Appendix B.

2.4 OIS Forward Rates

The MHSC validation analyzed the smoothness of forward curves (computed from zero-rates using the forward Libor formula), and observed following effects on OIS forward rates from MLib:

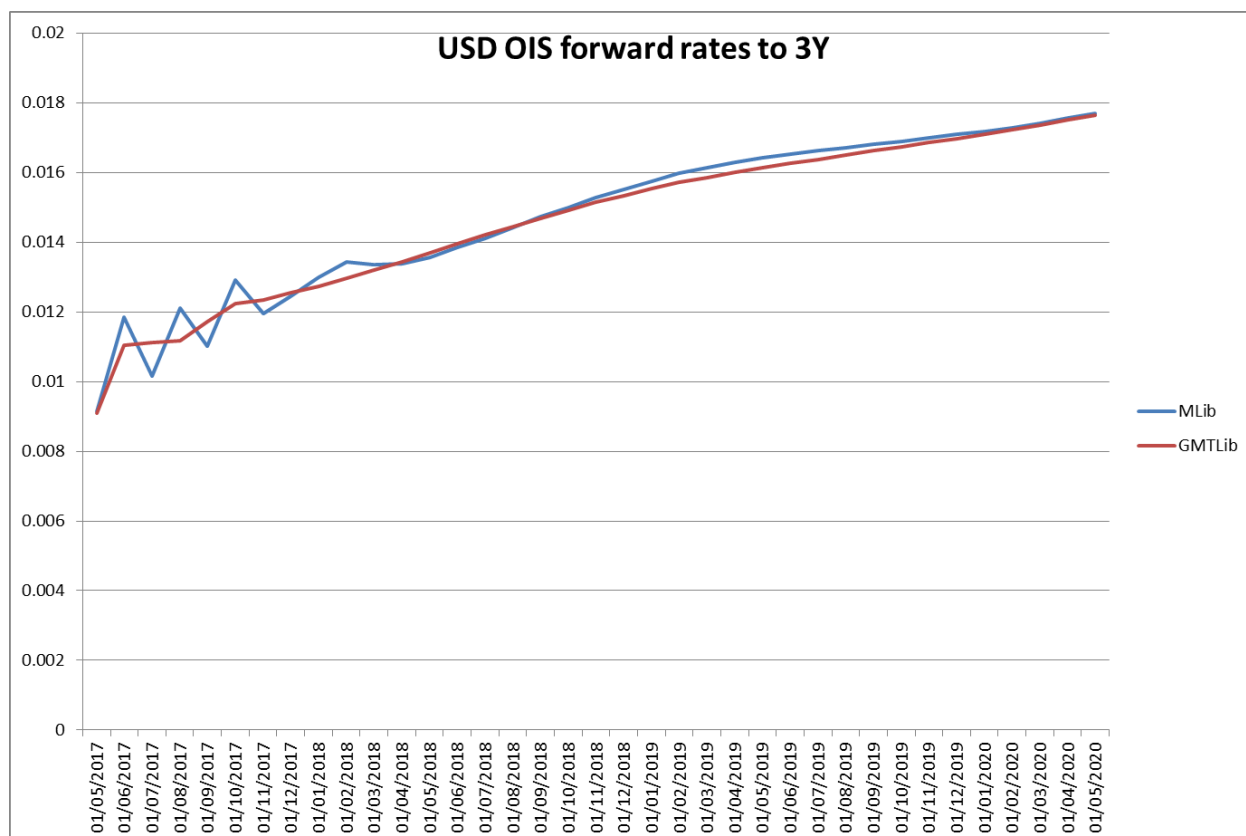
- Zig-zag on the short end
- Curtain effect on the mid / long end

Other curves (e.g. Libor, basis) did not display these effects. These effects did not show up when computed in the benchmark library (GMTLib), where curves were constructed from the quotes (fully alternative approach).

To investigate further, we sought to isolate the forward rate calculation from curve calibration. This approach allows us to investigate the forward rate calculation (easy), and the calibration (hard) in turn. We check the forward rates by starting with calibrated zero-rates on pillars from MLib, and then use them to compute forward rates independently (GMTLib). The zero-rates built from 23/05/2017 data are:

Term	Date	YF	DF	ZeroRate
ON	24/05/2017	0.0027397	0.9999747	0.009226272
1W	30/05/2017	0.0191781	0.9998251	0.009118168
2W	06/06/2017	0.0383562	0.9996483	0.009169633
3W	13/06/2017	0.0575342	0.9994691	0.009230531
1M	23/06/2017	0.0849315	0.9991807	0.009650648
2M	24/07/2017	0.169863	0.998222	0.010476678
3M	23/08/2017	0.2520548	0.9972985	0.010732456
4M	25/09/2017	0.3424658	0.9962563	0.010952016
5M	23/10/2017	0.4191781	0.9953229	0.011183874
6M	24/11/2017	0.5068493	0.9942374	0.0114024
9M	23/02/2018	0.7561644	0.9910679	0.01186542
1Y	23/05/2018	1	0.9878113	0.012263596
2Y	23/05/2019	2	0.9728581	0.013758534
3Y	26/05/2020	3.0109589	0.9561851	0.01488022
4Y	24/05/2021	4.0054795	0.9385169	0.015841897
5Y	23/05/2022	5.0027397	0.9199003	0.016688852
7Y	23/05/2024	7.0054795	0.8812948	0.018037746
10Y	24/05/2027	10.008219	0.8221944	0.019561758
12Y	23/05/2029	12.008219	0.7838805	0.020277672
15Y	24/05/2032	15.013699	0.7295231	0.021005096
20Y	26/05/2037	20.021918	0.6491807	0.021578564
25Y	23/05/2042	25.016438	0.5814376	0.021675809
30Y	23/05/2047	30.019178	0.5214366	0.021691719

We compare MLib and GMTLib forward rates:

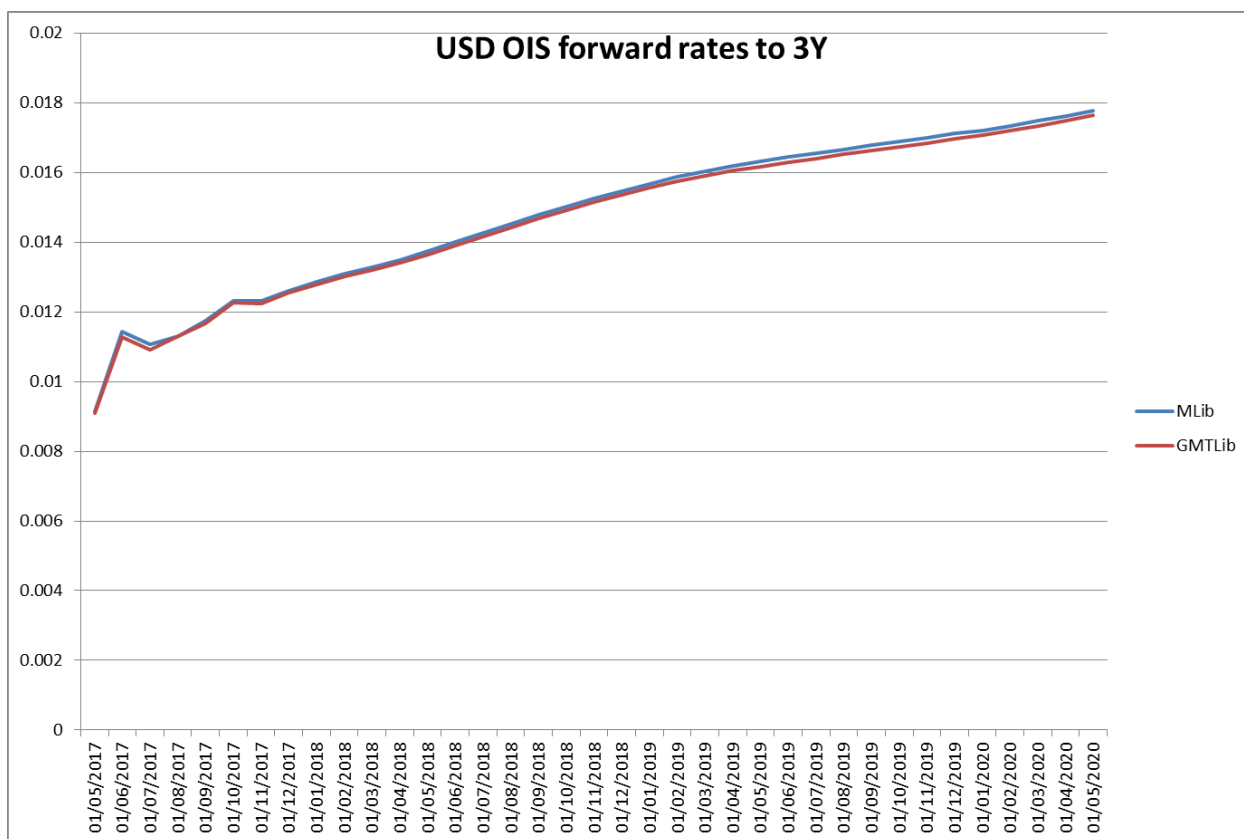


We find that that GMTLib has no zig-zag effect using the same zero-rates, while MLib does.⁷ This implies that calibration is not the cause of the zig-zag effect. The MLib forward rate calculation must be using information other than zero-rates on pillars.

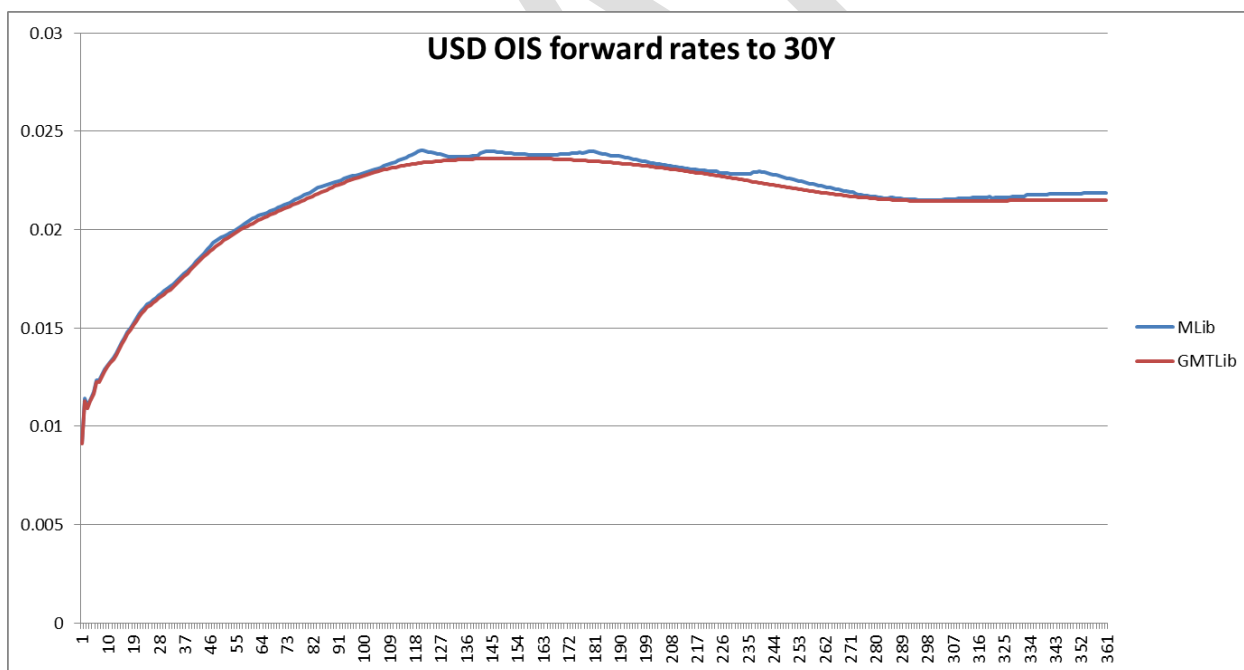
From discussion with model owners, we find this effect is caused by the curve setting *SmoothShortEnd* = *TRUE*. The setting introduces additional (daily) calibration instruments with the same quotes in the short end. These additional points break our assumption that zero-rates on pillars are sufficient to interpolate on the OIS curve. We note this feature is designed to improve behaviour when using Central Bank Swap quotes (e.g. EUROIS curve), since the central bank rate is not expected to move between committee meetings. When not using CBS quotes, the flag should be set to *FALSE*.

We repeat the experiment with *SmoothShortEnd* = *FALSE* and find the two libraries have similar behaviour:

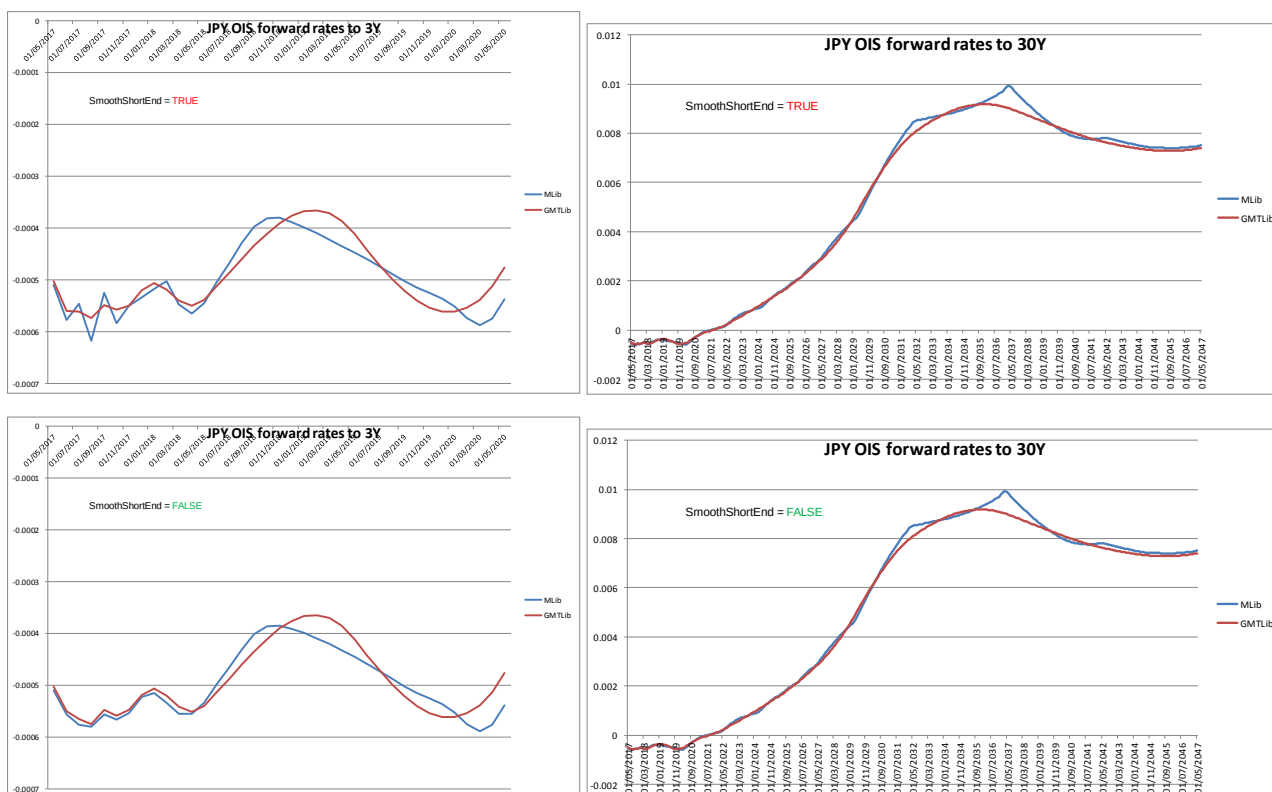
⁷ Here GMTLib forward rates are similar to those found in [MLibValMHSC17] when building from quotes in GMTLib.



This setting, however, does not resolve the curtain effect observed on the mid / long end of the curve:



The forward rates computed in MLib must therefore depend on something in addition to the zero-rates on pillar. Similar behaviour is observed on the JPY OIS forward rates:



This does not prove that anything is wrong with calibration, but that forward rates are not computed as expected, resulting in the curtain effect.

We received early feedback from the model owners that modifications to calibration can resolve the curtain effect, however this indicates that calibration & pricing (i.e. forward calculation) are inter-connected in ways that are not totally clear. This should be considered in more detail at the next review.

3 Documentation and Code

3.1 Document Sign Off

Date	Version	Author	Change Details
30/10/2017	1.0	Jeffrey Gilmore	Initial version.

3.2 Code and Validation Archive

All code, test tools, and results, can be found in

X:\Secure\Dir_Quantitative_Methodology\Approved Models\MLib

References

[MLibValMHSC17] MHSC, *Validation of Yield Curve Construction in M-Library*, Tokyo, 2017.

[MLib17] *MLib Yield Curves in a Nutshell*, London, 2017.

[MHIRM14a] MHI, Risk Management, *Validation of the Curve Construction in Murex*, 2014.

A Swap Template

Swap

LEGTTYPE	Leg1:Fixed	Leg2:Float
FORECASTCURVE		USD3ML
COMPOUNDMETHOD	NONE	NONE
DISCOUNTCURVE	USDOIS	USDOIS
VALUATIONCURRENCY	USD	USD
CURRENCY	USD	USD
NOTIONAL	1000000	1000000
NOTIONALEXCHANGE	NONE	NONE
LEVERAGE	1	1
PAYRECEIVE	PAY	RECEIVE
EFFECTIVEDATE	25/05/2017	25/05/2017
MATURITYDATE	25/05/2047	25/05/2047
ACCRUALFREQUENCY	Semi-Annual	Quarterly
ACCRUALBUSINESSDAYADJUSTMENT	MOD_FOLLOWING	MOD_FOLLOWING
ACCRUALCALENDAR	NYB:LNB	NYB:LNB
ACCRUALDAYCOUNT	30/360	ACT/360
PAYMENTFREQUENCY	Semi-Annual	Quarterly
PAYMENTBUSINESSDAYADJUSTMEN	MOD_FOLLOWING	MOD_FOLLOWING
PAYMENTCALENDAR	NYB:LNB	NYB:LNB
PAYMENTLAG	0D	0D
FIRSTSTUBDATE		
LASTSTUBDATE		
ROLLDAY	NORMAL	NORMAL
STUBTYPE	ShortStart	ShortStart
FIXINGBUSINESSDAYADJUSTMENT		MOD_FOLLOWING
FIXINGCALENDAR		NYB:LNB
FIXINGLAG		0D
FIXINGADVANCEORARREAR		
FIXEDRATE	2.4889%	
FLOATSPREAD		0
FIRSTFIXING		
LASTFIXING		
FIRSTSTUBCURVEINDEX		NATURAL
LASTSTUBCURVEINDEX		NATURAL

B Libor Fixing

The MLib Libor fixing procedure follows the algorithm below. Terms in **red** refer to floating leg accrual dates. Terms in **green** (with underscore) refer to Libor index start / end dates:

fixing = **accrual start**

accrual start = **accrual start**

accrual end = **accrual start** + index term (under Libor holiday / business day conventions)

Libor Day Count Conv = Leg Day Count Conv

ForwardLibor = $(P(\text{accrual start}) / P(\text{accrual end}) - 1) / \text{year_frac}(\text{accrual start}, \text{accrual end}, \text{Libor Day Count Conv})$

Coupon = ForwardLibor * year_frac(**accrual start**, **accrual end**, Leg Day Count Conv)